

TW15: Unsupervised Machine Learning

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose
Cluster customers, reduce dimensionality, and validate unsupervised findings without treating visual patterns as objective truth.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Prepare and justify the clustering space

Scenario

Customer features use pounds, counts, rates, and index values with very different scales.

INSTRUCTIONS

1. Import customer_segments.csv and hold customer_id and reference_segment outside fitting.
2. Audit missingness, distributions, and scale.
3. Standardise the four clustering features.
4. Verify transformed means and standard deviations.
5. Explain how feature selection defines the meaning of distance.

REQUIRED SUBMISSION

- A validated feature matrix and scaled matrix.
- Before-and-after scale summaries.
- A feature-choice rationale and limitation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Choose k using several forms of evidence

Scenario

The business expects a small number of usable segments but has not fixed k.

INSTRUCTIONS

1. Fit k-means for k=2 through 8 using n_init=20 and random_state 42.
2. Record inertia, silhouette, cluster sizes, and smallest-cluster share.
3. Create elbow and silhouette charts.
4. Recommend k using separation, stability, size, and interpretability.

REQUIRED SUBMISSION

- A complete k-comparison table.
- Two labelled evidence charts.
- A reasoned k recommendation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Profile k-means and hierarchical clusters

Scenario

Stakeholders need interpretable segment evidence and a comparison of two algorithms.

INSTRUCTIONS

1. Fit chosen k-means and Ward agglomerative clustering on the same scaled data.
2. Profile counts, medians, means, and variability in original units.
3. Draw a truncated dendrogram.
4. Compare assignments with adjusted Rand index.
5. Create neutral descriptive names and one possible action per segment.

REQUIRED SUBMISSION

- Two cluster assignments and one ARI value.
- Original-unit profiles and dendrogram.
- Non-judgemental segment descriptions with limitations.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Apply PCA and interpret loadings

Scenario

The four scaled features need a lower-dimensional summary for interpretation.

INSTRUCTIONS

1. Fit full PCA and report individual and cumulative explained variance.
2. Choose the smallest component count reaching at least 90 percent variance.
3. Create a loading table and identify the two largest absolute contributors to PC1 and PC2.
4. Plot the first two components coloured by cluster.
5. Explain arbitrary component sign and lost information.

REQUIRED SUBMISSION

- Explained-variance and loading tables.
- A two-component plot.
- A careful interpretation of components and limitations.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Validate t-SNE and cluster stability

Scenario

A visually separated embedding has been presented as proof of three customer types.

INSTRUCTIONS

1. Create t-SNE embeddings for perplexities 10, 30, and 40 with the same random seed.
2. Compare k-means solutions across at least five seeds using ARI and silhouette.
3. Use `reference_segment` only after fitting and report external ARI.
4. Explain what t-SNE preserves and what its global gaps cannot establish.
5. Write an unsupervised validation report and restricted intended use.

REQUIRED SUBMISSION

- Three t-SNE panels.
- Stability, separation, and external-reference tables.
- A validation report that avoids claiming objective customer truth.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____